Clusterformer for Pine Tree Disease Identification Based on UAV Remote Sensing Image Segmentation

Huan Liu, Wei Li, *Senior Member, IEEE*, Wen Jia, Hong Sun, Mengmeng Zhang, Lujie Song, and Yuanyuan Gui

***Abstract*— Pine wilt disease (PWD) is one of the most prevalent pine tree diseases, resulting in both ecological and economic havoc. Unmanned aerial vehicle (UAV) remote sensing segmentation plays a crucial role in early identifying and preventing PWD. However, deep learning segmentation models customized for PWD identification in scenarios with complex backgrounds have not received extensive exploration. In this article, we propose a novel UAV remote sensing segmentation model called Clusterformer with a conventional encoder–decoder structure. The encoder is comprised of the specially designed cluster transformer, which includes a cluster token mixer and a spatial-channel feed-forward network (SC-FFN). The cluster token mixer utilizes constructed clusters from the feature maps to represent pixels, thereby reducing redundant and interfering information. The SC-FFN extracts multiscale spatial information through depth-wise convolutions and channel information through a multilayer perceptron (MLP) in sequence. The decoder primarily consists of the specially designed D-cluster transformer. The token mixer of the D-cluster transformer employs constructed clusters from high-level decoded tokens to represent low-level encoded tokens without relying on traditional upsampling methods such as interpolation, transpose convolution, or patch expansion. Consequently, more robust and less redundant features from high-level decoded feature maps are transferred to low-level encoded feature maps. Experimental results on two PWD datasets demonstrate that Clusterformer outperforms existing state-of-the-art segmentation models. This confirms the effectiveness and efficiency of Clusterformer in PWD identification. The code is available at https://github.com/huanliu233/Clusterformer.**

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Huan Liu, Wei Li, Mengmeng Zhang, Lujie Song, and Yuanyuan Gui are with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: huanliu233@gmail.com; liwei089@ieee.org; mengmengzhang@bit.edu.cn; bufeisong0315@163.com; 3220205108@bit.edu.cn).

Wen Jia is with the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing 100091, China, and also with the Key Laboratory of Forestry Remote Sensing and Information System, National Forestry and Grassland Administration, Beijing 100091, China (e-mail: jiawen@ifrit.ac.cn).

Hong Sun is with the Center for Biological Disaster Prevention and Control, National Forestry and Grassland Administration, Shenyang 110034, China (e-mail: sunhongcaf@163.com).

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I. Introduction

PINE wilt disease (PWD) is one of the most severe pine tree diseases, colloquially referred to as the "cancer" of pine trees [1]. This malady is brought about by the pine wood nematode (PWN), scientifically known as *Bursaphelenchus xylophilus*. PWD has wrought considerable ecological and economic havoc, leading to the degradation of billions of pine trees [2], [3], [4]. In efforts to prevent the spread of PWD, extant strategies for PWD management primarily hinge on the early detection of afflicted trees and subsequent intervention methods, encompassing tree felling, incineration, and fumigation procedures [5]. Consequently, the imperative nature of conceiving an effective approach for PWD identification becomes apparent, underscored by the potential utilization of remote sensing techniques.

Unmanned aerial vehicle (UAV) remote sensing emerges as a resourceful and adaptable modality, particularly well-suited for scrutinizing environments characterized by modest to intermediate spatial extents. These aerial platforms can be equipped with remote sensing instruments designed to capture high-resolution imagery, thereby holding significant promise in the field of forest pest and disease detection. Within the realm of UAV remote sensing, researchers have developed numerous approaches for detecting PWD. For PWD detection, Syifa et al. [6] employed an artificial neural network (ANN) and support vector machine (SVM), successfully discriminating between healthy pine trees and those afflicted by PWD. Iordache et al. [7] collected multiple UAV remote sensing datasets and utilized the random forest (RF) algorithm to assess classification accuracy in the context of PWD identification. Similarly, Yu et al. [8] employed the RF algorithm to estimate the stage of PWD infection in trees sampled through UAV-based remote sensing, as well as data collected directly from ground-based tree assessments. Notably, these investigations relied on conventional machine learning methods, such as RF and SVM, for identifying PWD-infected trees. However, it should be emphasized that conventional machine learning methods do not possess the capability to directly discern specific objects within images.

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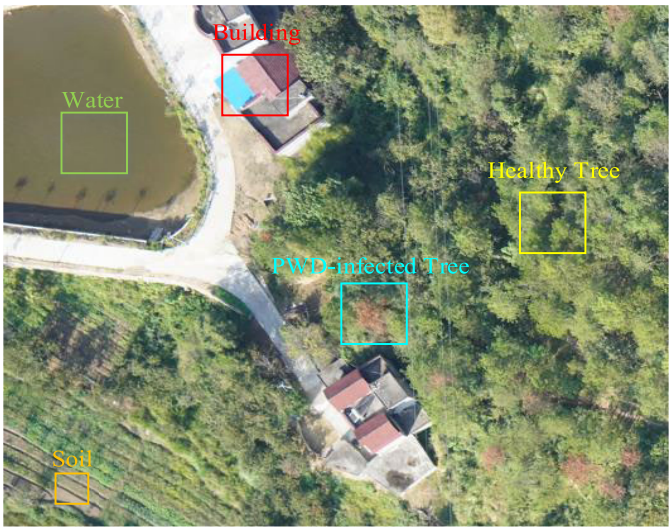


Fig. 1. Illustration of the complex background for PWD identification.

Compared to conventional machine learning methods, deep-learning-based approaches have demonstrated the capacity to extract more profound and discriminative features [9], [10], [11], [12], [13], [14], [15]. As deep learning has advanced, UAV-based PWD identification techniques have evolved through various methodologies. Yu et al. [8] introduced a target detection dataset for PWD identification, involving the use of bounding boxes to label PWD-infected trees. They employed established deep-learning-based target detection models, namely, faster R-CNN [16] and YOLOv4 [10], for the detection of PWD-infected trees. Xia et al. [17] emphasized that deep learning techniques offer the most efficient means of identifying infected pine trees, particularly for assessing the potential spread of PWD over expansive areas. They curated a segmentation dataset for PWD identification, where pixel-level labeling was applied to PWD-infected trees. The study encompassed an exploration of PWD segmentation performance utilizing established segmentation models, such as FCN [18], DeepLabv3+ [19], and PSPNet [9]. Similarly, Wang et al. [20] selected three deep-learning-based semantic segmentation models, namely, DeepLabv3+, HRNet [21], and DANet [22]. They integrated these models with generative adversarial networks (GANs) to identify discolored standing tree samples indicative of PWD in remote sensing images captured by the Gaofen-2 satellite.

However, the aforementioned studies predominantly relied on existing models without the development of deep learning models specifically tailored for PWD identification. From a semantic segmentation perspective, PWD identification presents a binary semantic segmentation challenge. In the context of UAV-based remote sensing imagery, the background encompasses a greater complexity, incorporating elements such as healthy trees, buildings, soil, and water, among others, as depicted in Fig. 1. Consequently, this environment introduces an abundance of redundant and interfering features, complicating accurate PWD segmentation. In addition, the different stages of PWD-infected trees exhibit different characteristics [8], which also confuses the models to identify the PWD-infected trees. Motivated by these challenges, we introduce Clusterformer, a UAV-based remote sensing semantic segmentation model designed for PWD identification. Clusterformer leverages cluster representations to mitigate redundancy and interference while extracting more robust features for PWD segmentation. The architecture of Clusterformer adheres to a conventional encoder–decoder structure. The encoder incorporates multistage feature extractors for different scales of feature maps, primarily driven by the proposed cluster transformer. Drawing inspiration from the well-known transformer [23], the cluster transformer comprises a cluster token mixer and a spatial-channel feed-forward network (SC-FFN) by adhering to the academic terminology in [24]. The cluster token mixer facilitates the transfer of information from constructed clusters within the feature map to individual pixels, representing them using cluster-based features, thereby enhancing pixel-level robustness. The SC-FFN is meticulously designed to extract multiscale spatial information through depth-wise convolution and channel information through a multilayer perceptron (MLP) in sequence. The decoder including the specially-designed D-cluster transformer likewise encompasses multistage feature decoding for different scales of feature maps, primarily driven by the decoding structure of the transformer. The SC-FFN structure in the D-cluster transformer mirrors that of the cluster transformer. The D-cluster token mixer is conceptually similar to that in the cluster transformer; however, it transfers cluster information from the high-level decoded feature maps to the low-level encoded feature maps, ensuring the extraction of more robust and less redundant information.

This article has the following significant contributions.

1) We present Clusterformer, a novel UAV-based remote sensing segmentation model for PWD identification. Clusterformer leverages cluster representations to effectively mitigate redundancy and interference, resulting in enhanced robustness in feature extraction for accurate PWD identification. Our model's superior performance has been rigorously validated against state-of-the-art models using datasets that are gathered and meticulously annotated by domain experts.

2) The cluster token mixer within the cluster transformer is specifically engineered to facilitate the transfer of constructed cluster information from the feature map to individual pixels, representing them using cluster-based features. Consequently, this approach yields more robust features for both PWD and non-PWD pixels. Furthermore, the designed SC-FFN within the cluster transformer adopts a structure that sequentially extracts multiscale spatial and channel features through depth-wise convolution and MLP, respectively, resulting in a more comprehensive feature extraction and fusion.

3) The D-cluster token mixer in the D-cluster transformer is tailored to transfer constructed cluster information from the high-level decoded feature map to pixels within the low-level encoded feature map, resulting in the extraction of more robust and less redundant features.

The remaining sections of this article are organized as follows. Section II presents the related works for segmentation models. Section III provides a detailed explanation of the proposed Clusterformer. Data description and experimental

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results are presented in Section IV. Finally, we conclude in Section V.

II. Related Works

A. CNN-Based Segmentation Models

Early deep learning models employed for semantic segmentation primarily revolved around convolutional neural networks (CNNs) [25], [26]. A pioneering example of such a CNN model is the fully convolutional network (FCN) introduced by Long et al. [18]. As the name suggests, FCN adheres to an encoder–decoder architecture, utilizing convolutional layers extensively in both the encoder and decoder while eschewing fully connected layers. This design choice results in a parameter-efficient model. Zhao et al. [9] advanced FCN by introducing the pyramid scene parsing network (PSPNet). PSPNet utilizes a multiscale pyramid pooling module (PPM) to extract diverse patterns from input images using a residual network (ResNet). These patterns are subsequently channeled through the PPM to capture both local and global context information. Recognizing that CNNs inherently operate within local neighborhood regions, which can limit their capacity to comprehend complex scenes, Zhao et al. [27] further proposed the point-wise spatial attention network (PSANet). PSANet addresses this limitation by aggregating information from other positions within an image through a self-adaptively learned attention mask. This approach enhances the model's predictive capabilities at the current position by incorporating information from other relevant areas.

One prominent lineage of image segmentation models comprises the Deeplab family [19], [28], [29], [30]. A seminal innovation within the Deeplab family is the introduction of atrous convolutions. This technique allows for the utilization of different dilation rates, enabling the acquisition of larger receptive fields for convolutions without the need for image resolution reduction through downsampling. Consequently, this approach facilitates the aggregation of more local information with expanded receptive fields, thereby enhancing pixel prediction capabilities. In addition to atrous convolutions, Deeplabv1 [28] incorporated fully connected conditional random fields (CRFs) to refine prediction maps. Subsequent versions of Deeplab, such as Deeplabv2 [29], improved segmentation performance through the introduction of various innovative modules and design enhancements. Deeplabv2 introduced atrous spatial pyramid pooling to leverage multiscale information, while Deeplabv3 [30] employed a multigrid strategy that incorporated multiple dilated convolutions. Furthermore, Deeplabv3+ adopted a UNet-like [31] structure, leveraging low-level feature maps that encapsulate more detailed information to further enhance segmentation performance.

Ronneberger et al. [31] introduced the UNet architecture in response to the observation that the decoder component of many semantic segmentation models underutilized low-level feature maps, resulting in the loss of crucial detailed information. UNet addresses this limitation by enhancing the FCN architecture through the incorporation of skip connections between corresponding low-level and high-level feature maps in the encoder and decoder. Subsequently, several extensions of UNet have emerged, all aiming to maximize the utilization of low-level feature maps [32], [33], [34]. The inclusion of skip connections has been recognized as a pivotal element in improving segmentation performance, as it enables the effective use of multiscale low-level feature maps. Beyond UNet, the feature pyramid network (FPN) [35], initially developed for object detection, introduced lateral connections to harness multiscale low-level feature maps. Consequently, both UNet and FPN, which leverage multiscale low-level feature maps, have exerted a significant influence on subsequent model designs. For instance, UPerNet [36] adopted an FPN-like structure, with the distinction of applying PPM to the highest-level feature maps while fusing multiscale feature maps.

B. Transformer-Based Segmentation Models

Following the groundbreaking work of Vaswani et al. [23] in the realm of natural language processing, researchers have embarked on the extension of transformer models into the domain of image processing. The vision transformer (ViT) [37] stands as the pioneering endeavor to introduce transformer architecture into image classification. ViT's architectural innovation commences with the tokenization of images, achieved by dividing them into nonoverlapping patches. Subsequently, ViT extracts features through the application of the multihead self-attention mechanism, as originally proposed in the transformer model. Distinct from CNN, which primarily captures local context information, ViT demonstrates an ability to capture long-range dependencies through the information exchange facilitated by the self-attention mechanism among all tokens [38], [39].

In the context of image segmentation models, the segmentation transformer (SETR) [40] marked a significant milestone by replacing CNNs with ViT as the encoder backbone. SETR's decoder employed either progressive upsampling or a multilevel aggregation feature strategy to derive the final probability map. Nevertheless, a noteworthy drawback of SETR was the substantial down-sampling rate, occurring exclusively during the tokenization process (e.g., 16). Such a high down-sampling rate proved less advantageous for dense prediction tasks such as segmentation. To address this issue, Xie et al. [41] introduced Segformer, which integrates a hierarchical structure with ViT to obtain multiscale feature maps from different stages. Within each stage, down-sampling is performed through the merging of overlapping patches, while feature extraction modules are composed of an efficient self-attention module and a mixed feed-forward network (mixed-FFN) as a variant of the transformer block. Other notable contributions, including MISSformer [42] and DAE-Former [43], have also embraced the hierarchical structure. MISSformer leverages a dual Transformer block, while DAE-Former incorporates an enhanced Transformer block as its basic unit. Both models integrate efficient self-attention modules as variants of the self-attention mechanism, aiming to optimize computational complexity.

SwinTransformer [44], a variant of the ViT, has demonstrated superior classification performance when compared to its predecessor ViT. One of its pivotal advancements lies

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in the utilization of a hierarchical structure, with particular emphasis on window- and shift window-based self-attention mechanisms. Building upon the Swin transformer block as its foundational unit, Cao et al. [45] introduced Swin-Unet, a model that embraces a UNet-like architecture and employs Swin transformer blocks for feature extraction. A noteworthy innovation within Swin-Unet is a patch expansion module for upsampling, a departure from conventional techniques such as interpolation and transpose convolution. Similarly, dc-Swin, proposed by Wang et al. [46], leverages the Swin transformer as its backbone and incorporates a dense feature aggregation module to extract multiscale relation-enhanced semantic features, thereby enhancing precision in segmentation tasks.

C. Hybrid CNN-and-Transformer-Based Segmentation Models

While segmentation models based on the ViT exhibit the capability to capture long-range dependencies, their performance tends to lag behind that of their CNN counterparts, especially when the training dataset is not sufficiently expansive. This performance disparity can be attributed to ViT's relative weakness in capturing local spatial information, which is a crucial aspect effectively leveraged by CNNs but is less emphasized in ViT. To compensate for this limitation and to harness local spatial information, a substantial volume of training data is often required.

To address the challenge of ViT's limited local representation, several methods have been proposed to construct hybrid CNN-transformer networks [47], [48]. Some approaches incorporate both CNN and transformer as backbone components. For instance, Chen et al. [49] introduced TransUNet, a model that employs CNN and transformer consecutively as backbones, with the output of the last CNN layer serving as input to the transformer. To exploit the multiscale feature maps generated by CNN in various stages, HiFormer [50] introduces the Swin transformer as a lateral backbone, allowing it to take in the outputs of CNN at corresponding stages. Similarly, Topformer [51] utilizes CNN as the initial backbone and incorporates multiple transformer blocks to collaboratively process feature maps at different stages of CNN, thereby reducing computational overhead. Some approaches adopt a hybrid CNN-and-Transformer block as their basic building unit. Grounded in the notion that CNN can be regarded as a high-frequency filter and transformer as a low-frequency filter, inception transformer (IFormer) [52] utilizes depthwise convolution, max-pooling, and transformer operations on distinct groups of input features. Similarly, AFFormer [53] employs an adaptive frequency filter as the fundamental unit, and adopts parallel transformer and convolution blocks to jointly process input features.

In the proposed Clusterformer, multiscale depthwise convolutions are used in the SC-FFN to aggregate the spatial information. In addition, convolution position embedding and relative convolution position embedding [54] are used to encode position information. Therefore, the Clusterformer can be regarded as one of the hybrid CNN-and-transformer-based segmentation models.

D. Segmentation Models for PWD Identification

The prosperity of semantic segmentation models has inspired the utilization of the existing semantic segmentation models for PWD identification. By incorporating deep learning semantic segmentation into PWD identification, Lee et al. [55] explored the performance of the existing segmentation models U-Net, SegNet, and DeepLab V3+ for PWD identification. Similarly, to thoroughly study the performance of the existing segmentation models for PWD identification, Xia et al. [17] adopted FCN, DeepLabv3+, and PSPNet on their collected datasets from a pine forest. By constructing a dataset of discolored standing tree samples of PWD from Gaofen-2 remote sensing images, Wang et al. [20] selected three semantic segmentation models—DeepLabv3+, HRNet, and DANet to compare their performance for PWD identification. To establish an effective method for PWD identification and extract infected regions, Wu and Jiang [56] adopted a mask R-CNN-based segmentation model for PWD identification. Li et al. [57] also utilized the mask R-CNN segmentation model on UAV-based hyperspectral images for PWD identification.

Different from the previous works that adopted the existing segmentation models, some works proposed their own segmentation models for PWD identification. Qin et al. [58] proposed a new network, spatial-context-attention network (SCANet), to identify pine nematode disease based on UAV-based multispectral remote sensing images. To identify the standing dead trees (SDTs) after PWD, Han et al. [59] proposed a multiscale spatial supervision convolutional network (MSSCN) to identify SDTs in a wide range of complex scenes based on UAV-based remote sensing images.

However, the works that used the existing models or constructed their own models for PWD identification did not consider that the PWD identification is in scenarios with complex backgrounds, and needs more robust representations to reduce the interference from the complex background. In the experimental part, it can be seen that these models only achieve relatively inferior segmentation performance, and the proposed Clusterformer that extracts more robust representations by utilizing constructing clusters outperforms these models with superior segmentation performance.

III. Methodology

A. Architecture Overview

The architecture of the proposed Clusterformer is illustrated in Fig. 2, encompassing the primary framework of Clusterformer itself, along with its constituent components: cluster transformer and D-cluster transformer. Clusterformer comprises an encoder and a decoder.

In the encoder module, UAV-based remote sensing images undergo an initial processing step within the patch embedding module. During this phase, the images are partitioned into nonoverlapping patches, each with a size of 4×4 pixels, subsequently transformed into tokens through a linear transformation. This process effectively downsamples the images by a factor of 4. The transformed patch tokens then traverse multiple cluster transformer blocks distributed across various feature extraction stages. These stages are interconnected

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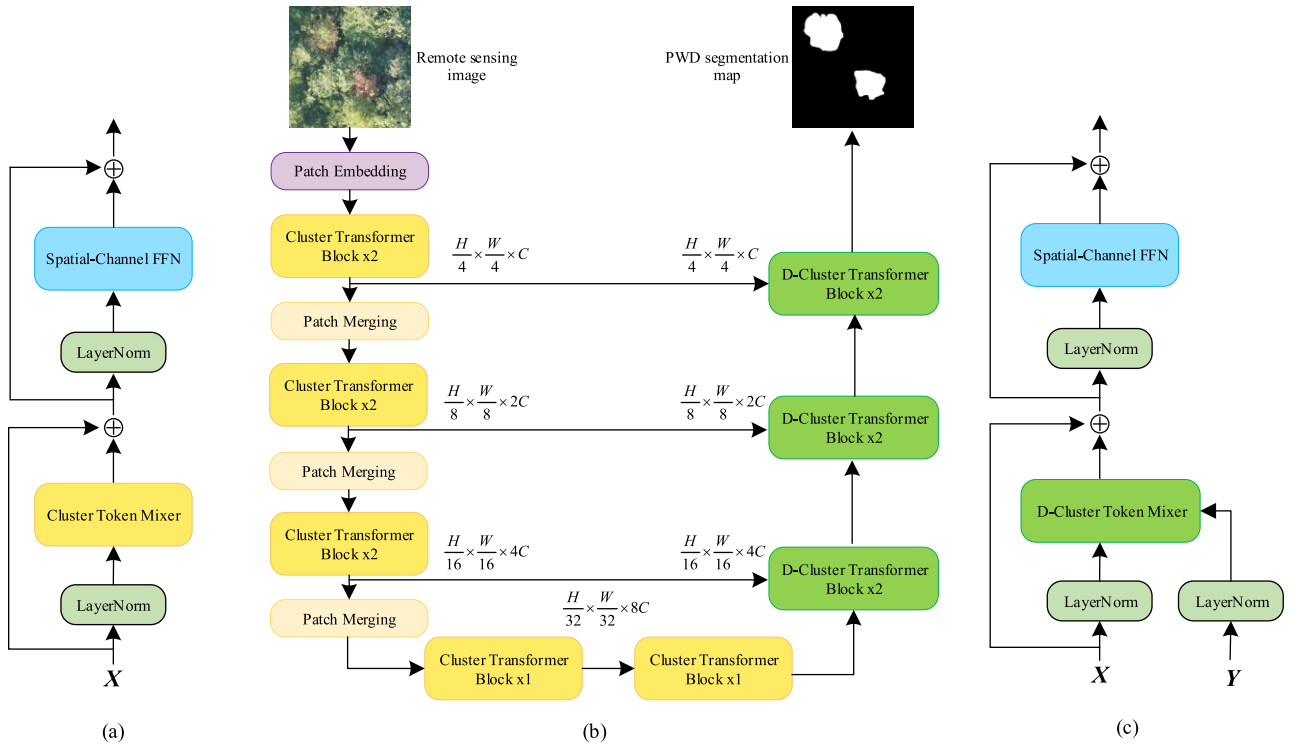


Fig. 2. Illustration of the proposed Clusterformer for PWD segmentation. (a) Cluster transformer. (b) Framework of Clusterformer. (c) D-cluster transformer.

by a patch merging module, resulting in the generation of hierarchical feature representations. Specifically, the cluster transformer block's responsibility is to acquire robust feature representations, while the patch merging module combines local patch tokens sized at 2×2 pixels, reducing the dimension of low-level feature maps by a factor of 2 through linear convolution. Note that the base channel dimension C in Fig. 2 is set as 64 in this article.

Inspired by the decoder structures of UNet [31] and FPN [35], the hierarchical decoder employs both low-level encoded tokens and high-level decoded tokens as inputs. However, unlike traditional implementations, this decoder does not rely on skip connections or lateral convolutions. Instead, the decoder is constructed using a D-cluster transformer, without resorting to conventional upsampling techniques such as interpolation, transpose convolution, or patch expansion as utilized in previous works [31], [45]. Drawing inspiration from cross-attention mechanisms in transformer decoders, the D-cluster transformer receives two inputs: one providing keys and values, and the other supplying queries. Note that in the D-cluster transformer, the high-level decoded tokens serve as keys and values, whereas the corresponding low-level encoded tokens serve as queries. This arrangement diverges from the typical transformer cross-attention setup. The motivation behind this design choice stems from the recognition that more discriminative information resides within the high-level decoded tokens, while detailed information is contained in the low-level encoded tokens. Consequently, the D-cluster transformer facilitates the transfer of values from high-level decoded tokens to low-level encoded tokens via cross-attention-like calculations. This maintains the same number of encoded tokens while preserving detailed information to the greatest extent possible. In essence, the D-cluster transformer serves as an upsampling module, effectively replacing the role of a separate upsampling module, such as interpolation, transpose convolution, or patch expansion. Further elaboration on the proposed Clusterformer is provided in the following.

B. Cluster Transformer

While the Vanilla transformer model excels at capturing long-range dependencies, thereby enabling the extraction of global information for feature extraction, it encounters several inherent limitations. Notably, it suffers from both quadratic computational complexity and a substantial requirement for training data to achieve promising results. These constraints restrict its practical utility, particularly when dealing with high-resolution images. In the context of PWD segmentation, the complexity of the background often surpasses that of the PWD-infected trees, leading to the presence of redundant and interfering information. In addition, different characteristics of the different stages of PWD-infected trees also hinder the models from accurate identification. In response to the need to mitigate these challenges, we propose the cluster transformer architecture, depicted in Fig. 3. The details of this innovative approach are elaborated subsequently.

1) Cluster Token Mixer: The central component of the cluster transformer is the cluster token mixer, akin to the self-attention mechanism introduced in the Transformer model by Vaswani et al. [23]. Suppose the feature maps and its corresponding tokens are $\mathbf{X}_m \in \mathbb{R}^{H \times W \times d}$ and $\mathbf{X} \in \mathbb{R}^{N \times d}$ ($N = H \times W$), respectively. By linear transformations, its

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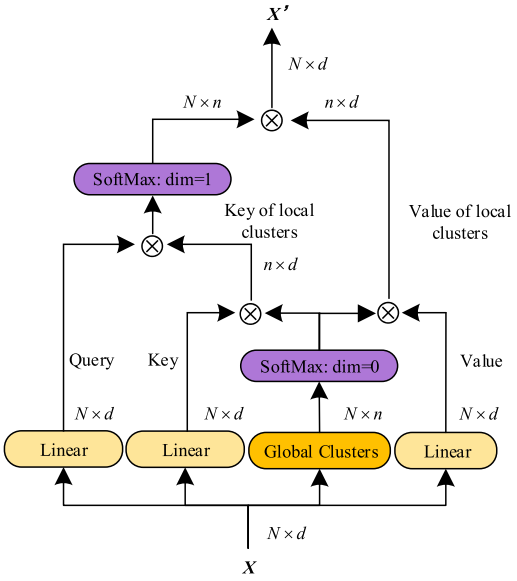


Fig. 3. Illustration of the proposed cluster token mixer.

query, key, and value are denoted as $\mathbf{Q} \in \mathbb{R}^{N\times d}$, $\mathbf{K} \in \mathbb{R}^{N\times d}$ and $\mathbf{V} \in \mathbb{R}^{N\times d}$, respectively. The standard self-attention calculation process is

$$A(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}, \text{dim} = 1\right)\mathbf{V}. \tag{1}$$

The computational complexity of this calculation is characterized by $O(H^2W^2d)$, a quadratic relationship with respect to the height and width of the feature maps. This quadratic complexity can be detrimental when dealing with high-resolution images. Given that PWD segmentation tasks often entail copious amounts of redundant and interfering information, there is a pressing need to simultaneously address both computational complexity and information redundancy. To tackle these challenges, we draw inspiration from established techniques in traditional image processing, such as K-means and superpixel segmentation, which utilize the concept of clusters to efficiently compress images. In our methodology, we define global clusters for the tokens across all image samples as $\mathbf{Z} \in \mathbb{R}^{n\times d}$, where n represents the number of clusters. The values and keys for the local clusters within the current image are, respectively,

$$\begin{aligned}\mathbf{K}_c &= \mathbf{K}\text{softmax}\left(\mathbf{X}\mathbf{Z}^T, \text{dim} = 0\right)\\ \mathbf{V}_c &= \mathbf{V}\text{softmax}\left(\mathbf{X}\mathbf{Z}^T, \text{dim} = 0\right)\end{aligned} \tag{2}$$

where $\text{softmax}\left(\mathbf{X}_t\mathbf{Z}^T, \text{dim} = 0\right)$ computes the similarity scores of all tokens within the current image sample with respect to the global clusters. Since the keys and values are based on the global clusters and are reconstructed from the local image, we called them the keys and values of the local clusters. Notably, different samples may exhibit distinct clusters. As expressed in (2), this process yields the key and value representations for the local clusters within the current image. This procedure bears a resemblance to the soft K-means algorithm utilized in updating the clusters. Subsequently, the cluster token mixer proceeds as follows:

$$C(\mathbf{Q}, \mathbf{K}_c, \mathbf{V}_c) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_c^T}{\sqrt{d}}, \text{dim} = 1\right)\mathbf{V}_c. \tag{3}$$

By referring to (2) and (3), we perform the computation of local clusters derived from global clusters. This operation facilitates the exchange of information between tokens and clusters, resulting in the acquisition of more robust token representations. Consequently, this approach effectively mitigates redundancy and interference within the image. Furthermore, the computational complexity of the cluster token mixer, as derived from (2) and (3), is denoted as $O(nHWd)$, a notably lower complexity compared to traditional self-attention mechanisms. This reduction in complexity is primarily attributed to the possibility of setting the number of clusters n to a value much smaller than HW. Moreover, when juxtaposed with renowned efficient attention mechanisms designed as variants of self-attention, engineered to alleviate computational complexity (typically characterized by a complexity of $O(HWd^2)$), our proposed cluster token mixer demonstrates competitive performance.

2) Multihead Cluster Token Mixer: Multihead self-attention has been shown to outperform single-head self-attention, as it allows tokens to capture information beyond their individual context, thereby mitigating overfitting. This principle aligns with the old saying, "Do not put all your eggs in one basket," emphasizing the importance of diversifying feature extraction to yield more robust representations. In a similar vein, we introduce a multihead cluster token mixer into our methodology. We denote the number of heads as h. Consequently, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are conventionally partitioned along the channel dimension, with each part residing in $\mathbb{R}^{N\times(d/h)}$. To further enhance the diversity of the cluster token mixer, we split $\mathbf{Z}$ along the first dimension, resulting in $\mathbf{Z}^1 \in \mathbb{R}^{n_1\times d}$, $\mathbf{Z}^2 \in \mathbb{R}^{n_2\times d}$, $\ldots$, $\mathbf{Z}^h \in \mathbb{R}^{n_h\times d}$. In essence, this means that the number of clusters differs for each head. The choice of the number of clusters is a crucial aspect of our method. The foundational value for this hyperparameter is determined to be proportional to the height or width of the corresponding feature map. This proportional relationship is established under the consideration that larger feature maps tend to accommodate more clusters. For distinct heads, the number of clusters associated with each is either half or twice the base number, contributing to a flexible and adaptive clustering configuration.

Subsequently, we apply $\mathbf{Q}^i$, $\mathbf{K}^i$, $\mathbf{V}^i$, and $\mathbf{Z}^i$ for the ith head ($1 \le i \le h$) to (2) and (3) for the cluster token mixer. The resulting outputs from all heads are then concatenated to yield the final output.

3) Position Embedding: In the ViT, the incorporation of spatial information into self-attention is achieved by inserting absolute position embeddings into the input data. Conversely, the Swin transformer adopts a different approach by introducing relative position embeddings to enhance the similarity calculations between tokens within windows. However, both of these methods are not well-suited for the proposed cluster token mixer, as the clusters themselves lack inherent spatial information. In cases where models leverage efficient attention mechanisms, such as those found in previous works like

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Dong et al. [53] and Xu et al. [54], where spatial information is not explicitly clear in the attention calculation, the introduction of convolution position embeddings and relative convolution position embeddings has been found to be effective in embedding spatial information. Drawing inspiration from these studies, we employ convolution position embeddings and relative convolution position embeddings to infuse spatial information into our cluster token mixer.

4) Spatial-Channel FFN: In the conventional transformer architecture, a feed-forward network (FFN) is typically connected to the self-attention mechanism to facilitate the extraction of nonlinear features. The conventional FFN (CFFN) adheres to a "linear-activation-linear" structural paradigm. However, this structure does not effectively leverage local spatial information. To enhance the correlation among neighboring tokens in the spatial dimension, Yuan et al. [60] introduced depth-wise convolution into the FFN, leading to the development of the locally-enhanced FFN (Le-FFN). The Le-FFN follows a "linear-activation-DWConv-activation-linear-activation" structure, which has demonstrated superior performance compared to the CFFN. Nonetheless, a challenge arises from the fact that the first linear transformation in Le-FFN typically expands the channel dimension by a certain ratio (e.g., 4). As a result, the depth-wise convolution in the subsequent layers introduces a significantly larger number of parameters and computational burden. In addition, the excessive use of activation and batch normalization layers in Le-FFN contributes to extra parameters and computational overhead.

To address these issues, we propose relocating the depth-wise convolution to the initial layer, followed by the application of a CFFN. This approach, named as the SC-FFN, offers dual advantages. First, it alleviates the computational burden compared to the Le-FFN architecture. Second, it enhances flexibility in the model. As the depth-wise convolution is directly connected to the multihead cluster token mixer, we can further enhance feature diversity by employing different kernel sizes in depth-wise convolution for different heads, as depicted in Fig. 4. In practice, we utilize larger kernel sizes for heads with a greater number of clusters and smaller kernel sizes for those with fewer clusters to strike a balance between detailed and global information.

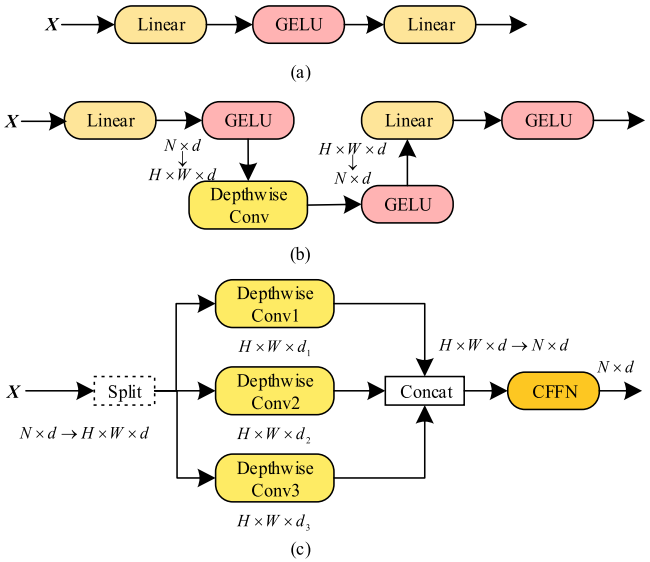


Fig. 4. Illustration of the proposed SC-FFN. (a) CFFN. (b) Le-FFN. (c) Proposed SC-FFN.

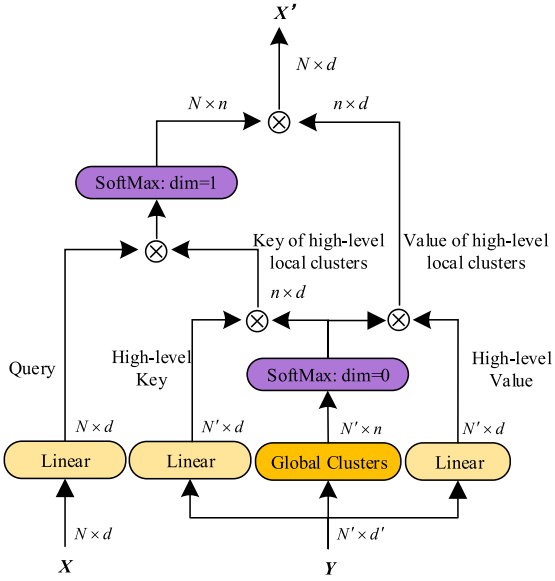


Fig. 5. Illustration of the proposed D-cluster token mixer.

C. D-Cluster Transformer

The conventional segmentation models typically employ an encoder–decoder architecture, wherein the decoder receives both encoded low-level feature maps and decoded high-level feature maps. This design ensures a balanced integration of detailed and discriminative information. Similarly, the Vanilla transformer architecture follows an encoder–decoder structure, where the decoder takes input in the form of queries derived from decoded tokens, as well as keys and values extracted from encoded tokens. Drawing inspiration from the Vanilla transformer, we introduce the D-cluster transformer for our decoder. The D-cluster transformer comprises two main components: the D-cluster token mixer and the SC-FFN which is also employed in the cluster transformer.

1) D-Cluster Token Mixer: Suppose the low-level encoded tokens and the corresponding high-level decoded tokens are $\mathbf{X} \in \mathbb{R}^{N\times d}$ and $\mathbf{Y} \in \mathbb{R}^{N'\times d'}$, respectively. As shown in Fig. 5, by linear transformations, the query of $\mathbf{X}$ and the key and value of $\mathbf{Y}$, are $\mathbf{Q} \in \mathbb{R}^{N\times d}$, $\mathbf{K}_{\text{de}} \in \mathbb{R}^{N'\times d}$, and $\mathbf{V}_{\text{de}} \in \mathbb{R}^{N'\times d}$, respectively. In addition, we introduce global clusters for the high-level decoded tokens across all samples, denoted as $\mathbf{Z}_{\text{de}} \in \mathbb{R}^{n\times d'}$, with n representing the number of clusters. The values and keys for the local clusters in the current sample are calculated as follows:

$$\begin{aligned}\mathbf{K}_{\text{dec}} &= \mathbf{K}_{\text{de}}\text{softmax}\left(\mathbf{Y}\mathbf{Z}_{\text{de}}^{T}, \text{dim} = 0\right)\\ \mathbf{V}_{\text{dec}} &= \mathbf{V}_{\text{de}}\text{softmax}\left(\mathbf{Y}\mathbf{Z}_{\text{de}}^{T}, \text{dim} = 0\right).\end{aligned} \tag{4}$$

Similar to (2), (4) gathers the key and value of the high-level decoded tokens or feature maps for the low-level encoded tokens. Then, the calculation process of the D-cluster token

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mixer is

$$dc(\mathbf{Q}, \mathbf{K}_{\text{dec}}, \mathbf{V}_{\text{dec}}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_{\text{dec}}^{T}}{\sqrt{d}}, \text{dim} = 1\right)\mathbf{V}_{\text{dec}}. \quad (5)$$

From (4) and (5), we derive the local clusters from the global clusters of the high-level decoded tokens, facilitating the low-level encoded tokens in gathering information from the high-level local clusters to obtain more robust representations. Consequently, more semantic information can be acquired without sacrificing detailed information. In addition, akin to the cluster token mixer, the computational complexity of the D-cluster token mixer is $O(nHWd)$, which is lower than that of self-attention and remains competitive with efficient and factorized attention.

2) Multihead D-Cluster Token Mixer: Similar to the cluster token mixer, we enhance the diversity of the D-cluster token mixer through a multihead mechanism. The transition from a single-head D-cluster token mixer to a multihead one resembles the multihead cluster token mixer. This involves splitting $\mathbf{Z}_{\text{de}}$ along the first dimension, $\mathbf{Q}$, $\mathbf{K}_{\text{de}}$, and $\mathbf{V}_{\text{de}}$ along the second dimension for all heads, and subsequently using their respective parts in the calculation of (4) and (5). Finally, the results from all heads are concatenated along the second dimension to yield the final output.

D. Training Objective

In the training of the Clusterformer, the conventional training objective typically involves entropy loss, defined as

$$\mathcal{L}_{en} = -\frac{1}{HW}\sum_{j=1}^{HW}\sum_{k=0}^{1}\log(P_{jk})\mathbf{1}(\text{label}_j = k) \quad (6)$$

where P_{ij} is the probability of the pixel i being in class j, and $\mathbf{1}$ is a logic function that outputs 1 if the input is true, or 0 if the input is false. We incorporate the entropy loss into our training objective. In addition, intersection over union (IoU) loss has been shown to be effective for training robust models as demonstrated in [61], and we include it in our training objective as well. The IoU is calculated as follows:

$$\mathcal{L}_{\text{IoU}} = -\frac{\sum_{j=1}^{HW} P_{ij}\mathbf{1}(\text{label}_j = 1)}{\sum_{j=1}^{HW} P_{ij} + \mathbf{1}(\text{label}_j = 1) - P_{ij}\mathbf{1}(\text{label}_j = 1)}. \quad (7)$$

By combining (6) and (7), the final training loss is

$$\mathcal{L} = \mathcal{L}_{en} + \lambda\mathcal{L}_{\text{IoU}} \quad (8)$$

where λ is a balance parameter that is fixed as 1e-1.

IV. Experimental Data and Results

A. Data Description

1) Hubei Dataset: The first dataset for PWD identification was collected by a Sony camera mounted on a UAV over a forest farm in the Hubei province of China. The original dataset consists of 64 remote sensing images in the RGB bands with the size of 6336 × 9504, and are meticulously annotated by domain experts for all stages of the PWD-infected trees. One sample of these remote sensing images and its corresponding label map is shown in Fig. 6. To fairly validate the proposed Clusterformer, 52 images are randomly selected as the training samples and the left 12 images as the testing samples. As the image sizes in the original dataset are too large, we crop these images into the images with the size of 512 × 512 by the stride 500. After the cropping, the number of training images is 12 844 while that of testing images is 2964.



Fig. 6. One sample of Hubei dataset for PWD identification and its corresponding label map.

2) Hunan Dataset: The second dataset for PWD identification was collected by a multispectral camera mounted on a UAV over a forest farm in the Hunan province of China. The original dataset consists of one remote sensing image in the RGB and infrared bands with the size of 14 262 × 8306, and are meticulously annotated by domain experts for all stages of the PWD-infected trees as shown in Fig. 7. We crop the image into the images with the size of 512 × 512 by the stride 500. After discarding the images with blank places, the number of randomly chosen training images is 183 while that of testing images is 80. Therefore, different from the Hubei dataset which consists of more image samples, the Hunan dataset has much fewer training and testing samples, which poses a greater challenge for testing model capacity in PWD identification.

B. Evaluation Metrics

Given that the primary segmentation target of this article is the identification of PWD-infected trees, our overall performance evaluation focuses primarily on PWD segmentation. Evaluating the overall segmentation performance, which includes both PWD and background, may lead to misleading results, especially since the number of PWD-infected samples is significantly smaller than that of the background. Consequently, all evaluation metrics are centered exclusively on PWD segmentation. We employ the following evaluation metrics: IoU, $F1$-score, precision, and recall for PWD. The

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TABLE I
SEGMENTATION PERFORMANCE AND MODEL COMPLEXITIES OF DIFFERENT MODELS FOR PWD IDENTIFICATION ON HUBEI DATASET

Group	Model	Encoder	Decoder	Recall (%)	Precision (%)	F1 (%)	IoU (%)	FLOPs (G)	Params (M)
CNN	UNet [31]	\	\	84.23	**89.51**	86.79	76.66	160.74	17.26
	ResUNet++ [34]	\	\	84.78	88.12	86.42	76.08	63.26	**4.06**
	PSPNet [9]	ResNet50	psphead	**87.13**	87.35	87.24	77.37	184.73	46.71
	PSANet [27]	ResNet50	psphead	85.4	**89.54**	**87.42**	**77.65**	199.44	53.16
	Deeplabv3+ [19]	ResNet50	\	84.97	**89.82**	**87.32**	**77.51**	173.40	39.76
	UperNet [36]	ResNet50	uperhead	**85.84**	88.83	87.31	77.49	42.52	31.34
	PIDNet [62]	\	\	85.48	88.61	87.02	77.02	22.25	28.54
	SCANet [58]	\	\	84.27	89.89	86.99	76.97	99.20	15.87
	MSSCN [59]	\	\	84.28	88.84	86.50	76.22	**13.05**	20.40
Transformer	SETR [40]	ViT-base	setr_up_head	81.96	84.52	84.52	71.26	91.76	86.66
	Segformer [41]	ViT-base	segformerhead	84.50	87.95	86.19	75.73	**13.16**	**7.72**
	SwinUNet [45]	SwinTransformer	SwinTransformer	81.63	88.59	84.97	73.87	40.93	41.39
	DAE-fomer [43]	\	\	85.63	89.38	**87.46**	**77.72**	105.59	48.07
Hybrid CNN and Transformer	TransUNet [49]	ResNet50+ViT	\	84.88	89.38	87.08	77.11	25.47	105.91
	iformer [52]	iformer-base	uperhead	84.32	**89.39**	86.78	76.65	230.77	40.80
	MISSFormer [42]	ViT-base	\	84.65	88.76	86.65	76.46	36.77	42.46
	AFFormer [53]	\	\	**86.41**	87.35	86.88	76.80	**4.79**	**2.90**
	Clusterformer	\	\	**87.06**	88.67	**87.85**	**78.35**	**17.69**	**11.34**



Fig. 7. One sample of Hunan dataset in the RGB bands for PWD identification and its corresponding label map.

calculation process for these metrics is as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{FN} + \text{FP} + \text{TP}} \tag{9}$$

$$F1 = \frac{2\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{FP} + \text{TP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{FN} + \text{TP}} \tag{12}$$

where true positive (TP) represents the number of the correctly predicted target pixels, false positive (FP) is the pixel number of the other classes that are falsely predicted to be the target, and false negative (FN) is the number of the falsely predicted target pixels.

C. Segmentation Performance

To validate the effectiveness of the proposed Clusterformer, experiments are conducted on the above-mentioned datasets in comparison with the state-of-the-art models, including CNN-based models, i.e., UNet [31], ResUNet++ [34], PSPNet [9], PSANet [27], Deeplabv3+ [19], UperNet [36], PIDNet [62], SCANet [58], and MSSCN [59], and transformer-based models, i.e., SETR [40], Segformer [41], SwinUNet [45], MISSFormer [42], and DAE-Transfomer [43], and hybrid CNN-and-Transformer-based models, i.e., TransUNet [49], iformer [52], and AFFormer [53].

1) Segmentation Accuracy: The segmentation results for various models, evaluated using different metrics on the dataset, are presented in Tables I and II. Results indicating relatively superior performance are highlighted in bold.

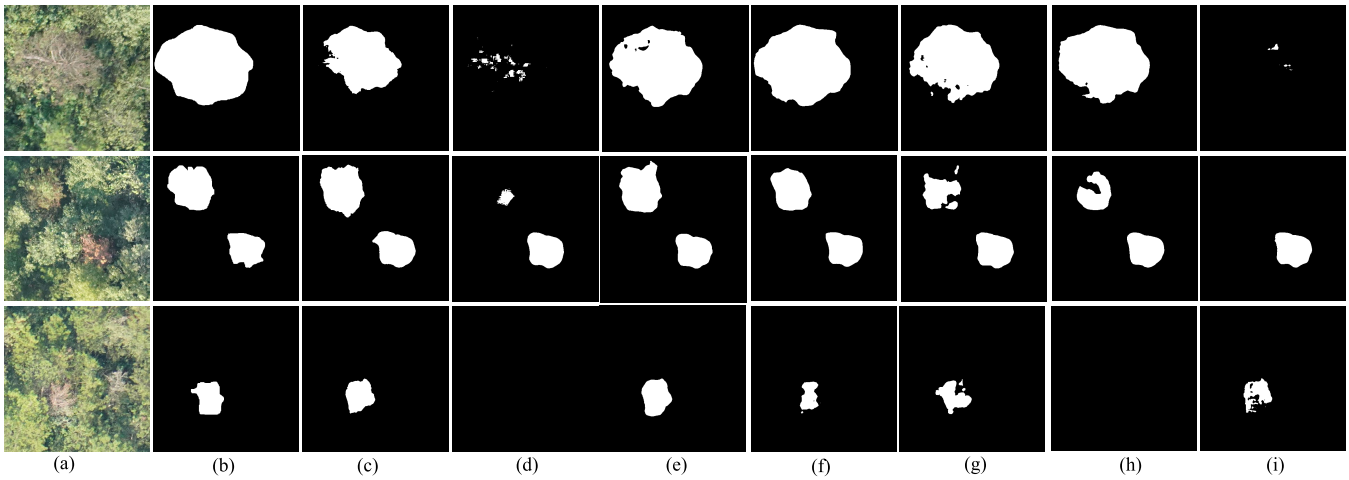
On the Hubei dataset, CNN segmentation models generally outperform transformer-based models. The effectiveness of CNN models can be attributed to their adept utilization of local information through spatial convolutions. However, among the transformer-based models, the DAE-former surpasses other transformer-based models as well as all CNN models in terms of segmentation performance. Hybrid CNN-and-transformer models, consistently outperform transformer-based models by incorporating convolutional layers or a CNN backbone into the transformer architecture. Furthermore, the proposed Clusterformer surpasses all compared models, boasting the highest performance. It outperforms the second-best model, DAE-former, by the margins of 0.63% on IoU and 0.39% on the $F1$ score.

On the Hunan dataset, some CNN segmentation models perform poorly, such as ResUNet++ and PIDNet, while

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TABLE II
SEGMENTATION PERFORMANCE [%] AND MODEL COMPLEXITIES OF DIFFERENT MODELS FOR PWD IDENTIFICATION ON HUNAN DATASET

Group	Model	Encoder	Decoder	Recall (%)	Precision (%)	F1 (%)	IoU (%)	FLOPs (G)	Params (M)
CNN	UNet [31]	\	\	75.65	71.92	73.74	58.40	160.90	17.26
	ResUNet++ [34]	\	\	**83.96**	38.41	52.71	35.78	63.30	**4.06**
	PSPNet [9]	ResNet50	PSPhead	71.40	68.88	70.12	54.00	184.77	46.71
	PSANet [27]	ResNet50	PSAhead	70.72	67.41	69.03	52.70	199.64	50.80
	Deeplabv3+ [19]	ResNet50	\	75.59	68.68	71.97	56.22	173.6	39.76
	UperNet [36]	ResNet50	uperhead	75.31	**74.53**	**74.91**	**59.89**	42.65	31.34
	PIDNet [62]	\	\	57.10	68.92	62.45	45.40	22.33	28.54
	SCANet [58]	\	\	72.74	**77.44**	**75.02**	**60.02**	99.40	15.88
	MSSCN [59]	\	\	67.53	72.11	69.75	53.55	**13.08**	20.40
Transformer	SETR [40]	ViT-base	setr_up_head	68.00	66.97	67.49	50.93	91.96	86.07
	Segformer [41]	ViT-base	segformerhead	72.23	71.98	72.10	56.38	**13.12**	**7.71**
	SwinUNet [45]	SwinTransformer	SwinTransformer	74.00	72.18	73.07	57.57	45.46	41.34
	DAE-former [43]	\	\	**76.47**	73.63	**75.02**	**60.03**	136.49	29.70
Hybrid CNN and Transformer	TransUNet [49]	ResNet50+ViT	\	73.42	70.72	72.05	56.31	129.25	93.23
	iformer [52]	iformer-base	uperhead	**76.99**	71.83	74.32	59.14	230.79	37.62
	MISSFormer [42]	ViT-base	\	65.41	**74.25**	69.55	53.32	37.05	35.46
	AFFormer [53]	\	\	68.84	66.00	67.38	50.81	**11.03**	**3.02**
	Clusterformer	\	\	**76.04**	**78.80**	**77.39**	**63.12**	**17.71**	**11.34**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 8. Remote sensing images, groundtruth, and segmentation maps of the proposed Clusterformer and other models that achieve above 77.10% of IoU on the Hubei dataset. (a) Remote sensing images. (b) Groundtruth. (c) Clusterformer (78.35%). (d) DAE-former (77.72%). (e) PSANet (77.65%). (f) Deeplabv3+ (77.51%). (g) UperNet (77.49%). (h) PSPNet (77.37%). (i) TransUNet (77.11%).

some CNN models perform consistently well, such as UperNet. Among the transformer-based models, the DAE-former still outperforms other transformer-based models as well as all CNN models. For hybrid CNN-and-transformer models, iformer and the proposed Clusterformer perform better than other hybrid CNN-and-transformer models. In particular, the proposed Clusterformer outperforms the second-best model, DAE-former, by the margins of 3.09% on IoU and 2.37% on the $F1$ score.

In terms of computational complexity, the proposed Clusterformer outperforms even the lightweight model PIDNet on both Hubei and Hunan datasets. It ranks third in terms of FLOPs and fourth in the number of parameters among all the compared models. This remarkable efficiency stems from the innovative cluster transformer and D-cluster transformer components, which substantially reduce computational demands through their cluster-based token mixing mechanisms. Overall, Clusterformer strikes an impressive balance between segmentation performance and computational complexity, emerging as the top-performing model among all compared alternatives.

2) Visualization: To visually illustrate the effectiveness of the proposed Clusterformer, segmentation maps generated by models with relatively high segmentation accuracy (IoU$\geq$77.10% and 57.50% on Hubei and Hunan datasets, respectively) are presented in Figs. 8 and 9. The segmentation results across all models are generally satisfactory for images where the PWD-infected tree is in the middle stage according to the categorization of different PWD infection stages in [8]. However, for images featuring PWD-infected trees in the early or late stages, some models exhibit deteriorating performance. For instance, on the Hubei dataset, both DAE-former and TransUNet struggle to segment the late-stage PWD-infected tree in the first image and the early-stage PWD-infected tree in the fifth image. This suggests that they are less proficient at identifying PWD-infected trees in these early and late stages.

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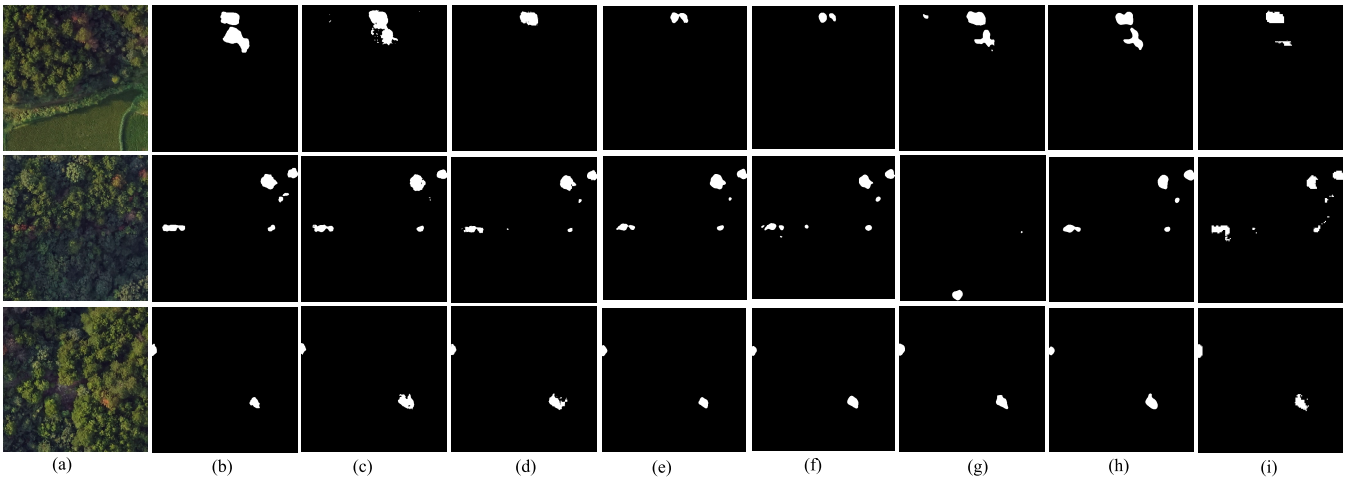


Fig. 9. Remote sensing images, groundtruth, and segmentation maps of the proposed Clusterformer and other models that achieve above 57.50% of IoU on the Hunan dataset. (a) Remote sensing images. (b) Groundtruth. (c) Clusterformer (63.12%). (d) DAE-former (60.03%). (e) SCANet (60.02%). (f) UperNet (59.89%). (g) Iformer (59.14%). (h) UNet (58.40%). (i) SwinUnet (57.57%).

TABLE III
ABLATION STUDY OF CLUSTER TOKEN MIXER AND THE DECODER WITH D-CLUSTER TOKEN MIXER ON HUBEI DATASET

Model	Recall (%)	Precision (%)	F1 (%)	IoU (%)	FLOPs (G)	Params (M)
Efficient attention + uperhead (EFU)	86.32	88.53	87.41	77.64	213.80	27.38
Efficient attention + our_decoder_head (EFD)	**89.36**	85.35	87.31	77.47	15.10	**11.06**
Factorised attention + uperhead (FAU)	86.58	88.32	87.44	77.69	213.73	27.37
Factorised attention + our_decoder_head (FAD)	84.41	**90.15**	87.19	77.29	**15.09**	11.08
Our attention + uperhead (CTU)	85.21	88.85	87.00	76.98	216.19	27.56
Our attention + our_decoder_head	87.06	88.67	**87.85**	**78.35**	17.69	11.34

TABLE IV
ABLATION STUDY OF CLUSTER TOKEN MIXER AND THE DECODER WITH D-CLUSTER TOKEN MIXER ON HUNAN DATASET

Model	Recall (%)	Precision (%)	F1 (%)	IoU (%)	FLOPs (G)	Params (M)
Efficient attention + uperhead (EFU)	71.65	77.02	74.24	59.03	213.81	27.38
Efficient attention + ourdecoder_head (EFD)	74.57	75.45	75.01	60.01	15.10	11.08
Factorised attention + uperhead (FAU)	66.78	78.27	72.07	56.34	213.81	27.38
Factorised attention + ourdecoder_head (FAD)	71.07	78.51	74.61	59.50	15.10	11.08
Our attention + uperhead (CTU)	71.66	77.55	74.49	59.35	216.21	27.56
Our attention + ourdecoder_head	76.04	78.80	77.39	63.12	17.71	11.34

In contrast, Clusterformer's performance remains relatively strong in the first and fifth images. Moreover, across all listed images, the segmentation maps produced by Clusterformer tend to be more accurate and exhibit smoother spatial transitions with fewer mislabeled pixels compared to the other models. These visual results align with the quantitative findings presented in Tables I and II, further corroborating the effectiveness of the proposed Clusterformer.

D. Ablation Study

To thoroughly evaluate the various components of the proposed Clusterformer, an ablation study is conducted. We compare the effectiveness of the proposed cluster token mixer with alternative attention mechanisms, specifically efficient attention, and factorized attention. Models corresponding to efficient attention and factorized attention are constructed by substituting the cluster token mixer with these alternative attention mechanisms in the encoder. Furthermore, we assess the impact of the proposed decoder structure, featuring the D-cluster transformer, by comparing it against the well-known uperhead. This comparison is executed by replacing the decoder with an uperhead in the corresponding models. Finally, the proposed SC-FFN is evaluated against other FFN configurations, namely, C-FFN and Le-FFN, to understand its contribution to the overall performance of the Clusterformer.

1) Ablation Study for the Cluster Token Mixer: The segmentation performances of the different models built during the ablation study, as evaluated using various metrics on Hubei and Hunan datasets, are presented in Tables III and IV, respectively. When replacing the cluster token mixer with either

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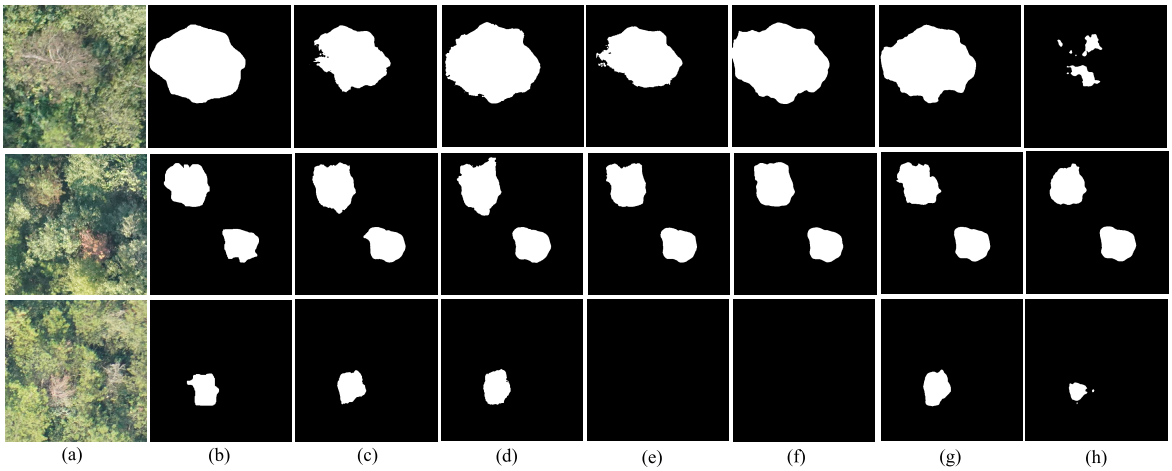


Fig. 10. Remote sensing images, groundtruth, and segmentation maps of the proposed Clusterformer and other built models with different attention modules and decoder structures on the Hubei dataset. (a) Remote sensing images. (b) Groundtruth. (c) Clusterformer (78.35%). (d) EFD (77.47%). (e) FAD (77.29%). (f) EFU (77.64%). (g) FAU (77.69%). (h) CTU (76.98%).

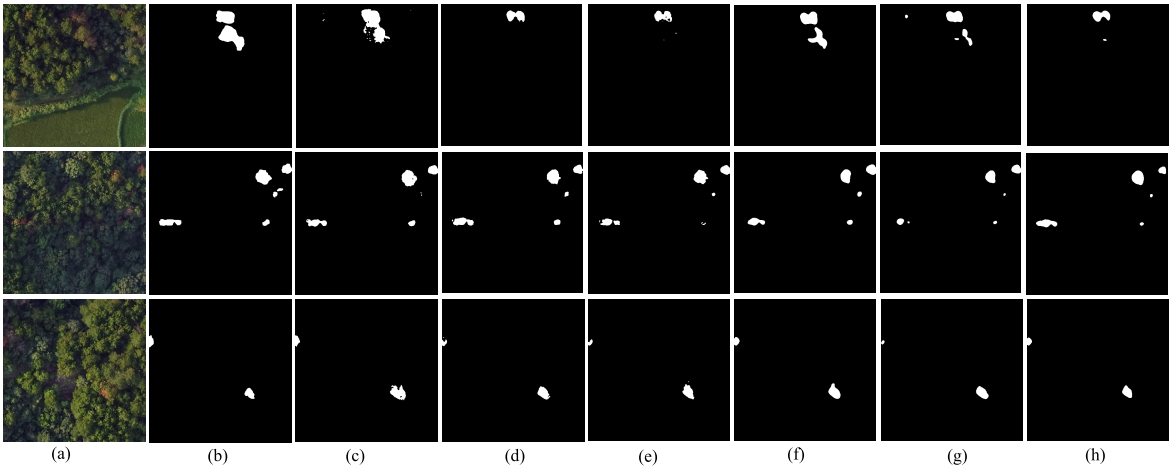


Fig. 11. Remote sensing images, groundtruth, and segmentation maps of the proposed Clusterformer and other built models with different attention modules and decoder structures on the Hubei dataset. (a) Remote sensing images. (b) Groundtruth. (c) Clusterformer (63.12%). (d) EFD (60.01%). (e) FAD (59.50%). (f) EFU (59.03%). (g) FAU (56.34%). (h) CTU (59.35%).

efficient attention or factorized attention, a significant decrease in segmentation performance is observed, with IoU scores dropping from 78.35% to 77.47% and 77.29%, and from 63.12% to 60.01% and 59.50% on Hubei and Hunan datasets, respectively. This stark contrast highlights the superior effectiveness of the cluster token mixer over these alternative attention mechanisms. Furthermore, the segmentation maps for these models are visualized in Figs. 10 and 11. Notably, the segmentation maps produced by the proposed Clusterformer consistently outperform those generated by models utilizing efficient attention and factorized attention, particularly when segmenting PWD-infected trees at different stages. These results serve as additional evidence supporting the effectiveness of the proposed cluster token mixer.

In terms of computational complexity, all three models are relatively comparable, with the proposed Clusterformer only slightly surpassing the other two in terms of FLOPs and the number of parameters. This reaffirms the efficiency of the cluster token mixer proposed in our model.

2) Ablation Study for SC-FFN: Tables V and VI display the segmentation results for C-FFN, Le-FFN, and the proposed SC-FFN on Hubei and Hunan datasets, respectively. Replacing the proposed SC-FFN with C-FFN or Le-FFN leads to a degradation in segmentation performance, with the IoU dropping from 78.35% to 77.48% or 77.63%, and from 63.12% to 58.84% or 59.78%, on Hubei and Hunan datasets, respectively. This provides clear evidence of the effectiveness of the SC-FFN. Furthermore, the segmentation results are visually illustrated in Figs. 12 and 13, which provides additional visual confirmation of the effectiveness of the proposed SC-FFN.

TABLE V
ABLATION STUDY OF THE SC-FFN ON HUBEI DATASET

Models	C-FFN	Le-FFN	SC-FFN
Recall (%)	86.45	86.47	**87.06**
Precision (%)	88.38	88.16	**88.67**
F1 (%)	87.41	87.31	**87.85**
IoU (%)	77.63	77.48	**78.35**
FLOPs (G)	**17.48**	17.77	17.69
Params (M)	**11.26**	11.37	11.34

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TABLE VI
ABLATION STUDY OF THE SC-FFN ON HUNAN DATASET

Models	C-FFN	Le-FFN	SC-FFN
Recall (%)	69.49	72.86	**76.04**
Precision (%)	79.34	76.90	**78.80**
F1 (%)	74.09	74.83	**77.39**
IoU (%)	58.84	59.78	**63.12**
FLOPs (G)	**17.50**	17.78	17.71
Params (M)	**11.26**	11.37	11.34

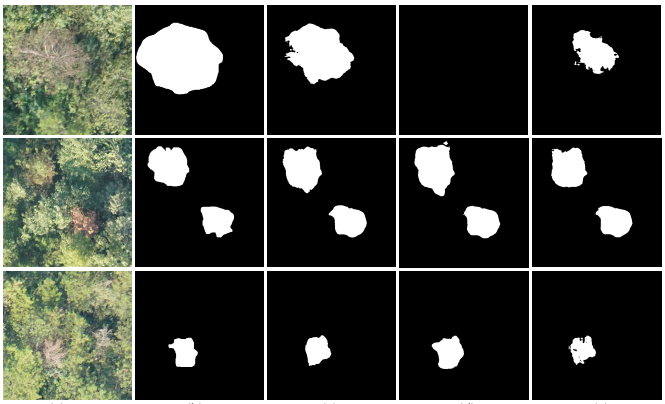


Fig. 12. Remote sensing images, groundtruth, and segmentation maps of the proposed Clusterformer and other built models with different FFNs on the Hubei dataset. (a) Remote sensing images. (b) Groundtruth. (c) Clusterformer (78.35%). (d) C-FFN (77.63%). (e) Le-FFN (77.48%).

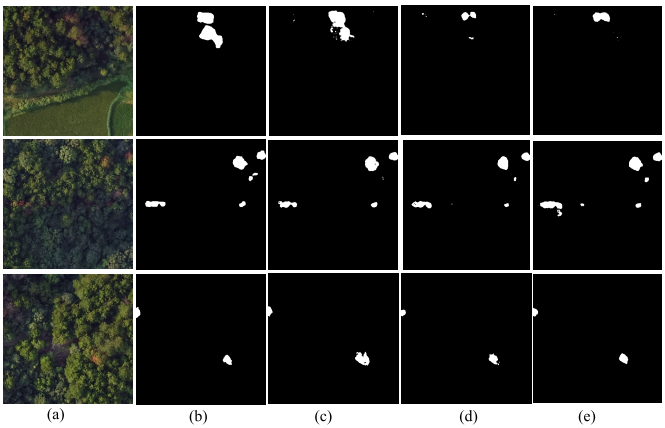


Fig. 13. Remote sensing images, groundtruth, and segmentation maps of the proposed Clusterformer and other built models with different FFNs on the Hunan dataset. (a) Remote sensing images. (b) Groundtruth. (c) Clusterformer (63.15%). (d) C-FFN (58.84%). (e) Le-FFN (59.78%).

In terms of computational complexity, the models built with C-FFN, Le-FFN, and the proposed SC-FFN are generally comparable. The proposed FC-FFN is slightly more complex than C-FFN but slightly less complex than Le-FFN, as evidenced by the FLOPs and the number of parameters. Therefore, the proposed FC-FFN significantly improves segmentation results while introducing only a negligible increase in computational complexity.

3) Ablation Study for the Proposed Decoder Structure With D-Cluster Transformer: Tables III and IV reveal that when we replace the decoder with uperhead for Clusterformer, there is a substantial degradation in segmentation performance, with the IoU dropping from 78.35% to 76.98%, and from 63.12%.to 59.35% on Hubei and Hunan datasets, respectively. This significant drop underscores the effectiveness of the proposed decoder structure with the D-cluster transformer. The segmentation maps presented in Figs. 10 and 11 reinforce these findings. Specifically, when comparing Clusterformer + uperhead (CTU) with efficient attention + uperhead (EFU) and factorized attention + uperhead (FAU), the maps generated by Clusterformer consistently outperform the other models in terms of segmenting PWD-infected trees at different stages. This further emphasizes the effectiveness of the proposed decoder structure.

However, it is noted that the proposed decoder structure does not exhibit a significant advantage over uperhead when the cluster token mixer is replaced with efficient attention or factorized attention on the Hubei dataset. The IoU for the models employing factorized attention or efficient attention with the proposed decoder structure are lower than those using uperhead. One possible explanation is that the encoders equipped with efficient attention and factorized attention may not effectively extract the robust features represented by clusters. Consequently, the proposed decoder structure struggles to decode well-represented clusters for the high-level decoded tokens and translate them into accurate segmentation of low-level encoded tokens.

For the computational complexity, from Tables III and IV, the models with the proposed decoder structures are all significantly less than those with uperhead in terms of the FLOPs and the number of parameters. This confirms the efficiency of the proposed decoder structure with a D-cluster transformer.

V. CONCLUSION

In this study, an effective and efficient UAV segmentation model, called Clusterformer, was proposed for PWD identification. The Clusterformer utilizes an encoder–decoder structure to identify and segment PWD-infected trees. The encoder mainly consists of the designed cluster transformer that can extract more robust, and less redundant and interfered information by utilizing the constructed clusters and well-positioned multiscale depth-wise convolutions. The decoder is mainly composed of the designed D-cluster transformer that uses the constructed clusters from the high-level decoded tokens to represent the low-level encoded tokens, transferring more robust and less redundant features from the high-level decoded feature maps to the low-level feature encoded maps. Experimental results on two PWD datasets have demonstrated the superior performance of the proposed Clusterformer, validating its effectiveness and efficiency in PWD identification.

REFERENCES

[1] H. Yu, H. Wu, X. Zhang, L. Wang, X. Zhang, and Y. Song, "Preliminary study on larix spp. infected by bursaphelenchus xylophilus in natural environment," *For. Pest Dis*, vol. 38, no. 4, pp. 7–10, 2019.

[2] C. Vicente, M. Espada, P. Vieira, and M. Mota, "Pine wilt disease: A threat to European forestry," *Eur. J. Plant Pathol.*, vol. 133, no. 1, pp. 89–99, May 2012.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:38:34 UTC from IEEE Xplore. Restrictions apply.

[3] M. Li, H. Li, R.-C. Sheng, H. Sun, S.-H. Sun, and F.-M. Chen, "The first record of *Monochamus saltuarius* (Coleoptera; Cerambycidae) as vector of *Bursaphelenchus xylophilus* and its new potential hosts in China," *Insects*, vol. 11, no. 9, p. 636, Sep. 2020.

[4] S. Mallez, C. Castagnone, E. Lombaert, P. Castagnone-Sereno, and T. Guillemaud, "Inference of the worldwide invasion routes of the pinewood nematode *Bursaphelenchus xylophilus* using approximate Bayesian computation analysis," *Peer Community J.*, vol. 1, p. e56, Dec. 2021.

[5] S.-R. Kim et al., "Hyperspectral analysis of pine wilt disease to determine an optimal detection index," *Forests*, vol. 9, no. 3, p. 115, Mar. 2018.

[6] M. Syifa, S.-J. Park, and C.-W. Lee, "Detection of the pine wilt disease tree candidates for drone remote sensing using artificial intelligence techniques," *Engineering*, vol. 6, no. 8, pp. 919–926, Aug. 2020.

[7] M.-D. Iordache, V. Mantas, E. Baltazar, K. Pauly, and N. Lewyckyj, "A machine learning approach to detecting pine wilt disease using airborne spectral imagery," *Remote Sens.*, vol. 12, no. 14, p. 2280, Jul. 2020.

[8] R. Yu, L. Ren, and Y. Luo, "Early detection of pine wilt disease in *Pinus tabuliformis* in north China using a field portable spectrometer and UAV-based hyperspectral imagery," *Forest Ecosyst.*, vol. 8, p. 44, Jan. 2021.

[9] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[10] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[11] W. Sun, K. Ren, X. Meng, G. Yang, J. Peng, and J. Li, "Unsupervised 3-D tensor subspace decomposition network for spatial–temporal–spectral fusion of hyperspectral and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528917, doi: 10.1109/TGRS.2023.3324028.

[12] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.

[13] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617515, doi: 10.1109/TGRS.2023.3302024.

[14] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600713, doi: 10.1109/TGRS.2022.3233726.

[15] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615, doi: 10.1109/TGRS.2023.3235819.

[16] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[17] L. Xia et al., "Evaluation of deep learning segmentation models for detection of pine wilt disease in unmanned aerial vehicle images," *Remote Sens.*, vol. 13, no. 18, p. 3594, Sep. 2021.

[18] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[19] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[20] J. Wang et al., "Satellite remote sensing identification of discolored standing trees for pine wilt disease based on semi-supervised deep learning," *Remote Sens.*, vol. 14, no. 23, p. 5936, Nov. 2022.

[21] K. Sun et al., "High-resolution representations for labeling pixels and regions," 2019, *arXiv:1904.04514*.

[22] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[23] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[24] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10819–10829.

[25] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.

[26] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5606216.

[27] H. Zhao et al., "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 267–283.

[28] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[29] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2017.

[30] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[31] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* (Lecture Notes in Computer Science), vol. 9351. Munich, Germany. Cham, Switzerland: Springer, 2015, pp. 234–241.

[32] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, Granada, Spain. Cham, Switzerland: Springer, 2018, pp. 3–11.

[33] H. Huang et al., "UNet 3+: A full-scale connected UNet for medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 1055–1059.

[34] D. Jha et al., "ResUNet++: An advanced architecture for medical image segmentation," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, Dec. 2019, pp. 225–2255.

[35] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[36] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 418–434.

[37] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[38] H.-F. Zhong, Q. Sun, H.-M. Sun, and R.-S. Jia, "NT-Net: A semantic segmentation network for extracting lake water bodies from optical remote sensing images based on transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627513.

[39] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607713.

[40] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[41] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[42] X. Huang, Z. Deng, D. Li, and X. Yuan, "MISSFormer: An effective medical image segmentation transformer," 2021, *arXiv:2109.07162*.

[43] R. Azad, R. Arimond, E. K. Aghdam, A. Kazerouni, and D. Merhof, "DAE-former: Dual attention-guided efficient transformer for medical image segmentation," 2022, *arXiv:2212.13504*.

[44] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[45] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, Tel Aviv, Israel. Cham, Switzerland: Springer, Oct. 2022, pp. 205–218.

[46] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[47] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

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[48] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, p. 1, 2022, Art. no. 4408715.

[49] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[50] M. Heidari et al., "HiFormer: Hierarchical multi-scale representations using transformers for medical image segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6191–6201.

[51] W. Zhang et al., "TopFormer: Token pyramid transformer for mobile semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12083–12093.

[52] C. Si, W. Yu, P. Zhou, Y. Zhou, X. Wang, and S. Yan, "Inception transformer," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 23495–23509.

[53] B. Dong, P. Wang, and F. Wang, "Head-free lightweight semantic segmentation with linear transformer," 2023, *arXiv:2301.04648*.

[54] W. Xu, Y. Xu, T. Chang, and Z. Tu, "Co-scale conv-attentional image transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9961–9970.

[55] M.-G. Lee, H.-B. Cho, S.-K. Youm, and S.-W. Kim, "Detection of pine wilt disease using time series UAV imagery and deep learning semantic segmentation," *Forests*, vol. 14, no. 8, p. 1576, Aug. 2023.

[56] Z. Wu and X. Jiang, "Extraction of pine wilt disease regions using UAV RGB imagery and improved mask R-CNN models fused with ConvNeXt," *Forests*, vol. 14, no. 8, p. 1672, Aug. 2023.

[57] H. Li, L. Chen, Z. Yao, N. Li, L. Long, and X. Zhang, "Intelligent identification of pine wilt disease infected individual trees using UAV-based hyperspectral imagery," *Remote Sens.*, vol. 15, no. 13, p. 3295, Jun. 2023.

[58] J. Qin, B. Wang, Y. Wu, Q. Lu, and H. Zhu, "Identifying pine wood nematode disease using UAV images and deep learning algorithms," *Remote Sens.*, vol. 13, no. 2, p. 162, Jan. 2021.

[59] Z. Han et al., "Detection of standing dead trees after pine wilt disease outbreak with airborne remote sensing imagery by multi-scale spatial attention deep learning and Gaussian kernel approach," *Remote Sens.*, vol. 14, no. 13, p. 3075, Jun. 2022.

[60] K. Yuan, S. Guo, Z. Liu, A. Zhou, F. Yu, and W. Wu, "Incorporating convolution designs into visual transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 579–588.

[61] B. Zhao et al., "Multiple attention network for spartina alterniflora segmentation using multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402915, doi: 10.1109/TGRS.2023.3272978.

[62] J. Xu, Z. Xiong, and S. P. Bhattacharyya, "PIDNet: A real-time semantic segmentation network inspired by PID controllers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19529–19539.

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